INTELLI CODE ANALYZER

AI Technical Debt Analysis & Optimization Comparison Report

Original Code Analysis | AI-Powered Refactoring | Before & After Results

KEY PERFORMANCE SNAPSHOT

Debt Score (Before)	Debt Score (After)	Improvement %	Issues Reduced (delta)
175	154	12.0%	10
Original scan	Optimized rescan	Score-based	Before minus after

Total Issues	Files Optimized	Parse Success %	Total Workflow Time
67 to 57	1	92.3%	52.88 s
Before to after	Modified source files	Before 100.0%	Measured components

Files discovered 13 to 13	**Parsed files** 13 to 12	**Parse failures** 0 to 1
Project size 15.4 KB to 14.7 KB	**Lines of code** 387 to 362	**Original scan** 0.24 s
AI generation 51.82 s	**Optimized rescan** 0.46 s	**Critical issues** 7.0 to 6.0
High issues 5.0 to 5.0	**Code smells** 67 to 57	**Security findings** 0 to 0

All values are populated from actual scan and optimization outputs. Debt-score improvement and issue-count reduction are separate measures.

Report Overview

This report compares the original and AI-optimized scans for **SampleProject** using the **java** analyzer profile. It summarizes measured findings, source statistics, optimization output, and file-level changes.

Executive Summary

SampleProject was analyzed as a **java** project. The baseline contained 67 issues with a debt score of 175.0. AI-assisted optimization processed the selected findings and the updated project was rescanned, resulting in 57 issues and a debt score of 154.0. The measured score change was -21.0 points; the comparison reports 10 fixed and 0 new issue(s). These are independent measures and should be reviewed alongside the changed source files.

Key Metric	Previous Code	Current Code	Change
Debt score	175.0	154.0	-21.0
Total issues	67	57	-10
Critical	7	6	-1
High	5	5	0
Medium	10	9	-1
Low	45	37	-8
Code smells	67	57	See category counts
Security findings	0	0	See category counts
Files modified	0	1	1

Project Overview

The original project is identified as SampleProject. The selected analyzer profile is java; stack-aware measurements below are taken from the scan's corresponding metrics object when available.

Attribute	Original Project	Optimized Project
Project name	SampleProject	SampleProject_20261104_122330_732168
Stack / language	java	java
Build configuration	maven	maven

Attribute	Original Project	Optimized Project
Files discovered	13	13
Files parsed	13	12
Parse failures	0	1
Project size	15.4 KB	14.7 KB
Classes / interfaces	12 / 1	11 / 1
Methods	62	61

Codebase Statistics

Source line and file-size measures are collected from stack-supported source files; unsupported or unavailable values are reported as N/A.

Metric	Before	After	Change
Files parsed	13	12	-1
Lines of code	387	362	-25
Comment lines	32	32	0
Blank lines	74	74	0
Classes	12	11	-1
Interfaces	1	1	0
Methods	62	61	-1
Project size	15.4 KB	14.7 KB	See values
Files modified	0	1	1
Changed lines (added / removed)	N/A	1 / 26	Generated file diffs

Code Smell Analysis

Code-smell counts include non-security source-analysis findings. The scan reported 67 such finding(s) before and 57 after optimization.

Rule	Severity	Before	After	Status
CircularDependency	high	1	1	Unchanged
DeadMethod	medium	1	1	Unchanged
DeepNesting	medium	1	1	Unchanged
DuplicateCode	medium	2	2	Unchanged
EmptyCatchBlock	high	1	1	Unchanged
ExcessiveParameterList	medium	2	2	Unchanged
FeatureEnvy	medium	2	1	Improved
GenericExceptionHandling	low	1	1	Unchanged
GodClass	critical	1	1	Unchanged
HardcodedCredentials	critical	2	2	Unchanged
HighCyclomaticComplexity	critical	3	2	Improved
LongMethod	high	1	1	Unchanged
MagicNumber	low	23	16	Improved
MissingJavadoc	low	20	19	Improved
MutableStaticField	high	1	1	Unchanged
StringConcatenationInLoop	medium	2	2	Unchanged
UnclosedResource	critical	2	2	Unchanged
UnusedVariable	low	1	1	Unchanged

4.1 Representative Issue Example

Rule: HighCyclomaticComplexity; severity: critical; location: src/demo/InvoiceService.java. Method 'calculateCustomerDiscount' has a Cyclomatic Complexity of 17 (threshold: 10).

Technical Debt Analysis

Debt score is read from the scan report. Score delta is current minus baseline; improvement is (baseline score - current score) / baseline score × 100 when baseline is positive. Issue-count delta is current issues minus baseline issues.

Severity	Before	After	Delta
Critical	7	6	-1
High	5	5	0
Medium	10	9	-1
Low	45	37	-8
TOTAL	67	57	-10

Code Quality and Maintainability

Quality metrics are derived from parsed class and method objects. Stack-specific metrics: Classes / methods: 12 / 62; Average / maximum cyclomatic complexity: 2.44 / 23.00; Dependency edges / cycles: 5 / 1.

Metric	Before	After	Interpretation
Cyclomatic complexity (average)	2.44	2.20	Lower is generally preferable
Cyclomatic complexity (maximum)	23.00	23.00	Lower is generally preferable
Methods above complexity threshold (10)	3.00	2.00	Lower is generally preferable
Average class fan-out	N/A	N/A	N/A: metric not emitted by this stack
Duplication percentage	N/A	N/A	N/A: metric not emitted by this stack

Dependency Analysis

Dependency totals and outdated-package counts use the dependency scanner output where available. Node.js package counts come from the Node stack analyzer. N/A means the scan did not provide that measure.

Metric	Before	After
Build configuration	maven	maven
Total dependencies	5	5
Outdated dependencies	2	2
Vulnerable / high-risk dependencies	2	2

Security Analysis

Source-code security findings are counted from issue categories. Dependency vulnerability totals are shown separately so source rules and package advisories are not conflated.

Source finding severity	Before	After
Critical	0	0
High	0	0
Medium	0	0
Low	0	0
Dependency advisories	2	2
Outdated dependencies	2	2

AI Optimization Summary

AI optimization results include completed generation attempts, modified files, and per-issue failures. A generated file is not proof that every associated issue was resolved; the optimized rescan is the validation source.

Metric	Measured value
Configured provider / model	gemini / gemini-3.1-flash-lite
Issues selected	2
Files processed / modified	2 / 1

Metric	Measured value
Failed issue generations	1
AI generation duration	51.82 s
Optimized rescan duration	0.46 s

File	Issue Addressed	Result
src/demo/InvoiceService.java	Selected issue(s)	Generated
K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java	HighCyclomaticComplexity	Failed: AI code generation failed: Gemini rate limit exceeded: You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. * Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 20, model: gemini-3.6-flash, description: Generate content requests per day limit exceeded on Free Tier. See https://ai.google.dev/gemini-api/docs/rate-limits. Please retry in 1h5m36.705214062s.. Please wait a moment before retrying.

Before and After Issue Comparison

Fixed means an issue key present in the baseline is absent in the optimized scan. New means the issue key appears only after optimization. Still present means a matching issue remains. Improved class counts are based on total findings per class.

Status	Count
Fixed	10
New	0
Still present	57
Improved classes	1
Worsened classes	0

Sample fixed issues

ID	Rule	Severity	File / location	Status
N/A	HighCyclomaticComplexity	critical	src/demo/InvoiceService.java:8	Fixed
N/A	MagicNumber	low	src/demo/InvoiceService.java:17	Fixed
N/A	MagicNumber	low	src/demo/InvoiceService.java:19	Fixed
N/A	MagicNumber	low	src/demo/InvoiceService.java:20	Fixed
N/A	MagicNumber	low	src/demo/InvoiceService.java:23	Fixed
N/A	MagicNumber	low	src/demo/InvoiceService.java:25	Fixed
N/A	MagicNumber	low	src/demo/InvoiceService.java:28	Fixed
N/A	MagicNumber	low	src/demo/InvoiceService.java:29	Fixed
N/A	FeatureEnvy	medium	src/demo/InvoiceService.java:8	Fixed
N/A	MissingJavadoc	low	src/demo/InvoiceService.java:8	Fixed

Sample new issues

ID	Rule	Severity	File / location	Status
N/A	N/A	N/A	N/A	No issues

Sample still-present issues

ID	Rule	Severity	File / location	Status
N/A	LongMethod	high	src/demo/UserService.java:21	Still present
N/A	ExcessiveParameterList	medium	src/demo/UserService.java:63	Still present
N/A	ExcessiveParameterList	medium	src/demo/UserService.java:90	Still present
N/A	GodClass	critical	src/demo/OrderProcessor.java	Still present
N/A	DuplicateCode	medium	src/demo/UserService.java:127	Still present
N/A	DuplicateCode	medium	src/demo/UserService.java:140	Still present
N/A	UnusedVariable	low	src/demo/UserService.java:22	Still present
N/A	DeadMethod	medium	src/demo/UserService.java:123	Still present
N/A	HighCyclomaticComplexity	high	src/demo/UserService.java:21	Still present
N/A	HighCyclomaticComplexity	critical	src/demo/UserService.java:63	Still present

Analysis and Optimization Efficiency

Workflow time is the sum of measured original scan, AI generation, optimized rescan, and comparison durations; it excludes user wait time outside those operations.

Efficiency metric	Value
Original scan duration	0.24 s
Files discovered (before / after)	13 / 13
Files parsed (before / after)	13 / 12
Parse success (before / after)	100.0% / 92.3%
AI generation duration	51.82 s
Optimized rescan duration	0.46 s
Comparison duration	0.13 s
Total workflow duration	52.88 s

File-Level Change Summary

File sizes and issue counts are shown for files returned by the optimization service. The generated result is based on the output source content.

File	Size Before	Size After	Issues Before	Issues After	Result
src/demo/InvoiceService.java	914 B	179 B	10	0	Updated

Class and Method Analysis

Class and method rows aggregate findings from analyzer issue records. Stacks without class-based parser results are represented by the analyzer's component/file names.

Class / component	Before findings	After findings	Status
demo.AuditLogService	2	2	Unchanged
demo.AuthManager	1	1	Unchanged
demo.Customer	2	2	Unchanged
demo.Employee	3	3	Unchanged
demo.FileStorageService	4	4	Unchanged
demo.GlobalConfig	2	2	Unchanged
demo.InvoiceService	10	0	Improved
demo.OrderProcessor	2	2	Unchanged
demo.PaymentGateway	3	3	Unchanged
demo.Person	1	1	Unchanged
demo.ReportGenerator	2	2	Unchanged

Class / component	Before findings	After findings	Status
demo.UserService	35	35	Unchanged

Reliability and Detection Coverage

Parse success describes successfully parsed files relative to discovered source files, not defect-detection accuracy. Precision, recall, and F1 require a verified ground-truth issue set and are therefore reported as N/A.

Coverage metric	Before	After
Files discovered	13	13
Files parsed	13	12
Parse success	100.0%	92.3%
Parse failures	0	1
Precision / recall / F1	N/A	N/A

Recommendations

- Resolve parse failures and rerun the analysis to improve source coverage.
- Review outdated dependencies and confirm compatible safe-version upgrades.
- Review failed AI-generation entries; manually inspect those files before release.

Final Summary

The figures below summarize the measured comparison. AI-generated code should be reviewed and validated before use.

Final metric	Result
Score change (after - before)	-21.0
Score improvement	12.0%
Issue-count change (after - before)	-10
Fixed / new / still present	10 / 0 / 57
Files modified	1
Status	Review generated changes and run project validation

Detailed Issue Appendix

The appendix lists issues returned by the baseline and optimized scans. Production use should include full issue descriptions, remediation guidance, and code context where available.

ID	Rule	Severity	File / location	Status
N/A	LongMethod	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:21	Current finding
N/A	ExcessiveParameterList	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:63	Current finding
N/A	ExcessiveParameterList	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:90	Current finding
N/A	GodClass	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\OrderProcessor.java	Current finding
N/A	DuplicateCode	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:127	Current finding

ID	Rule	Severity	File / location	Status
N/A	DuplicateCode	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:140	Current finding
N/A	UnusedVariable	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:22	Current finding
N/A	DeadMethod	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:123	Current finding
N/A	HighCyclomaticComplexity	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:8	Current finding
N/A	HighCyclomaticComplexity	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:21	Current finding
N/A	HighCyclomaticComplexity	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:63	Current finding
N/A	DeepNesting	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:95	Current finding
N/A	EmptyCatchBlock	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:110	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\Customer.java:37	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\Employee.java:18	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\FileStorageService.java:13	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:17	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:19	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:20	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:23	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:25	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:28	Current finding

ID	Rule	Severity	File / location	Status
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:29	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\PaymentGateway.java:13	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:27	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:78	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:78	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:79	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:79	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:84	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:87	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:87	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:97	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:120	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:131	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:144	Current finding
N/A	HardcodedCredentials	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:59	Current finding
N/A	HardcodedCredentials	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:59	Current finding
N/A	FeatureEnvy	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:8	Current finding

ID	Rule	Severity	File / location	Status
N/A	FeatureEnvy	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:63	Current finding
N/A	CircularDependency	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\AuditLogService.java:1	Current finding
N/A	StringConcatenationInLoop	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\Employee.java:37	Current finding
N/A	StringConcatenationInLoop	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\ReportGenerator.java:11	Current finding
N/A	UnclosedResource	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\FileStorageService.java:12	Current finding
N/A	UnclosedResource	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\FileStorageService.java:12	Current finding
N/A	MutableStaticField	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\GlobalConfig.java:11	Current finding
N/A	GenericExceptionHandling	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\PaymentGateway.java:8	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\AuditLogService.java:10	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\AuthManager.java:8	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\Customer.java:36	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\Employee.java:30	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\FileStorageService.java:11	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\GlobalConfig.java:13	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\InvoiceService.java:8	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\OrderProcessor.java:85	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\PaymentGateway.java:8	Current finding

ID	Rule	Severity	File / location	Status
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\Person.java:16	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\ReportGenerator.java:8	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:21	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:58	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:63	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:90	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:95	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:110	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:127	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:140	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\uploads\SampleProject\src\demo\UserService.java:161	Current finding
N/A	LongMethod	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:21	Current finding
N/A	ExcessiveParameterList	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:63	Current finding
N/A	ExcessiveParameterList	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:90	Current finding
N/A	GodClass	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\OrderProcessor.java	Current finding
N/A	DuplicateCode	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:127	Current finding
N/A	DuplicateCode	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:140	Current finding

ID	Rule	Severity	File / location	Status
N/A	UnusedVariable	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:22	Current finding
N/A	DeadMethod	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:123	Current finding
N/A	HighCyclomaticComplexity	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:21	Current finding
N/A	HighCyclomaticComplexity	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:63	Current finding
N/A	DeepNesting	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:95	Current finding
N/A	EmptyCatchBlock	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:110	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\Customer.java:37	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\Employee.java:18	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\FileStorageService.java:13	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\PaymentGateway.java:13	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:27	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:78	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:78	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:79	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:79	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:84	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:87	Current finding

ID	Rule	Severity	File / location	Status
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:87	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:97	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:120	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:131	Current finding
N/A	MagicNumber	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:144	Current finding
N/A	HardcodedCredentials	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:59	Current finding
N/A	HardcodedCredentials	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:59	Current finding
N/A	FeatureEnvy	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:63	Current finding
N/A	CircularDependency	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\AuditLogService.java:1	Current finding
N/A	StringConcatenationInLoop	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\Employee.java:37	Current finding
N/A	StringConcatenationInLoop	medium	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\ReportGenerator.java:11	Current finding
N/A	UnclosedResource	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\FileStorageService.java:12	Current finding
N/A	UnclosedResource	critical	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\FileStorageService.java:12	Current finding
N/A	MutableStaticField	high	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\GlobalConfig.java:11	Current finding
N/A	GenericExceptionHandling	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\PaymentGateway.java:8	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\AuditLogService.java:10	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\AuthManager.java:8	Current finding

ID	Rule	Severity	File / location	Status
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\Customer.java:36	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\Employee.java:30	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\FileStorageService.java:11	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\GlobalConfig.java:13	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\OrderProcessor.java:85	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\PaymentGateway.java:8	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\Person.java:16	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\ReportGenerator.java:8	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:21	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:58	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:63	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:90	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:95	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:110	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:127	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:140	Current finding
N/A	MissingJavadoc	low	K:\PROJECTS\MINI PROJECT\original_mini_project(technical_debt)\original mini project\AI-Technical-Debt-Agent\backend\scanner\output\optimized\SampleProject_20261104_122330_732168\src\demo\UserService.java:161	Current finding

Report Definitions and Notes

Previous code means the baseline scan. **Current code** means the optimized rescan. **Delta** is current minus baseline unless labeled otherwise. **Improvement percentage** is (baseline score - current score) / baseline score × 100 for a positive baseline score. **Fixed** means a baseline issue key is no longer present in the optimized scan. **N/A** indicates the metric was not available in the supplied scan output; it is not estimated. AI-generated changes are suggestions and require human review, tests, and deployment-specific validation.